Mel-Frequency Cepstral Coefficients for Speech Analysis

Mathematical Foundations, Python Implementation, and Feature Use

Technical Report

Abstract

This report explains Mel-Frequency Cepstral Coefficients (MFCCs), a widely used representation of short-time speech spectra. It develops the processing chain from waveform framing and the short-time Fourier transform through Mel filterbank energies, logarithmic compression, and the discrete cosine transform. The report explains the parameters in a librosa implementation, interprets the resulting matrix, discusses recording-level aggregation, and identifies considerations for using MFCCs in Parkinson's disease speech-classification research.

Contents

1 Introduction

Speech is a time-varying signal. Its spectral shape changes with phonetic content, articulation, vocal-tract configuration, and voice production. MFCCs provide a compact representation of the short-time spectral envelope. They are commonly used in speech recognition and speaker/audio classification because they reduce a high-dimensional spectrum to a smaller set of coefficients.

MFCCs are not direct measures of vocal-fold cycle irregularity. They describe spectral characteristics and therefore complement features such as jitter, shimmer, HNR, and timing measurements.

2 MFCC processing pipeline

The conventional MFCC pipeline contains the following stages:

1. Divide the waveform into short, usually overlapping frames.
2. Apply a window to reduce frame-boundary discontinuities.
3. Calculate a short-time Fourier transform (STFT).
4. Convert spectral power into Mel-filterbank energies.
5. Apply a logarithm to compress the dynamic range.
6. Apply a discrete cosine transform (DCT) to obtain cepstral coefficients.
7. Retain a selected number of coefficients.

3 Short-time framing and Fourier analysis

Let $x[n]$ be a discrete speech waveform. For frame index m, frame length L, and hop size H, a frame can be represented as

$$x_m[n] = x[n + mH]w[n], \qquad 0 \leq n < L,$$

where $w[n]$ is a window function. The STFT is

$$X(m, k) = \sum_{n=0}^{L-1} x[n + mH]w[n]e^{-j2\pi kn/N_{\mathrm{FFT}}}.$$

The magnitude or power spectrum describes how signal energy is distributed over frequency within each frame.

Framing is needed because speech is nonstationary. A single Fourier transform of an entire utterance would mix together different phonetic events and hide their temporal evolution. Short-time analysis provides a sequence of spectral descriptions.

4 The Mel frequency scale

The Mel scale is a nonlinear frequency mapping intended to approximate aspects of human pitch perception. One commonly used mapping is

$$m(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right).$$

Mel filterbanks use triangular filters distributed on this perceptual scale. More filters are concentrated in lower-frequency regions, while the spacing in hertz becomes wider at higher frequencies.

If $P_m[k]$ is the power spectrum of frame m and $H_r[k]$ is the rth Mel filter, the corresponding filterbank energy is conceptually

$$E_m[r] = \sum_k P_m[k] H_r[k].$$

This converts a large set of FFT bins into a smaller set of perceptually spaced band energies.

5 Logarithmic compression and DCT

The Mel energies are compressed using a logarithm:

$$L_m[r] = \log(E_m[r] + \epsilon),$$

where ϵ prevents taking the logarithm of zero. The discrete cosine transform then decorrelates and compacts the log-Mel representation:

$$c_m[q] = \sum_{r=0}^{M-1} L_m[r] \cos\left[\frac{\pi q}{M}\left(r + \frac{1}{2}\right)\right],$$

up to the normalization convention used by the implementation. Here M is the number of Mel bands and q is the coefficient index.

The first coefficient is strongly related to the average log spectral level. Subsequent coefficients describe increasingly fine variations in the log-Mel spectral shape. Retaining only a subset reduces dimensionality and can suppress some fine-scale variation.

6 Python implementation

The discussed code is:

```
mfcc = librosa.feature.mfcc(
    y=y,
    sr=sr,
    n_mfcc=13,
    n_fft=2048,
    hop_length=512,
    n_mels=40
)
```

6.1 Input waveform: `y=y`

The variable `y` is the sampled mono waveform. It should be a one-dimensional numerical array. MFCC extraction does not itself guarantee that the waveform has been normalized or resampled; those operations must be decided and documented separately.

6.2 Sampling rate: `sr=sr`

The sampling rate specifies how many samples represent one second. It is required to map FFT bins to physical frequencies and to construct the Mel filterbank. If the waveform was loaded at 16 kHz, then `sr` must be 16000. Passing an incorrect sampling rate changes the interpretation of the spectrum.

6.3 Number of coefficients: `n_mfcc=13`

This requests 13 coefficients per frame. Thirteen is a conventional baseline in speech processing, not a universal optimum. Lower-order coefficients tend to represent broad spectral-envelope properties, while higher-order coefficients capture finer variation. The appropriate number should be selected using the task, dataset size, and validation performance.

6.4 FFT size: `n_fft=2048`

The FFT size determines the number of frequency samples in the transform. At a sampling rate of 16 kHz, 2048 samples correspond to

$$\frac{2048}{16000} = 0.128 \text{ s} = 128 \text{ ms}.$$

In librosa, unless `win_length` is specified, the analysis window length defaults to `n_fft`. Thus this configuration uses a relatively long analysis window for speech. A longer window gives finer frequency resolution but reduces temporal localization. For rapid speech changes, a shorter window may be preferable. The choice should match the experimental protocol.

6.5 Hop length: `hop_length=512`

The hop is the number of samples by which the analysis window advances. At 16 kHz,

$$\frac{512}{16000} = 0.032 \text{ s} = 32 \text{ ms}.$$

The hop controls the spacing between successive MFCC vectors. It does not itself set the FFT frequency resolution. Since the default window is 2048 samples, adjacent windows overlap substantially.

6.6 Number of Mel bands: `n_mels=40`

This requests 40 Mel filterbank bands. The filterbank compresses the FFT spectrum into 40 band-energy values before logarithmic compression and the DCT. More bands provide a more detailed spectral representation but also increase the dimensionality of the intermediate representation.

7 Understanding the output

For a single waveform, librosa generally returns an array with shape

$$(n_{\text{mfcc}}, N_{\text{frames}}).$$

For 13 coefficients and 100 frames, the shape is $(13, 100)$. Each column corresponds to a frame, and each row corresponds to one coefficient index. Therefore, this code does not return 13 recording-level scalar features; it returns 13 time sequences.

8 Recording-level aggregation

If the classifier expects one feature vector per recording, a consistent aggregation step is required. For coefficient q, one may calculate

$$\mu_q = \frac{1}{T}\sum_{t=1}^{T} c_q(t), \qquad \sigma_q = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(c_q(t) - \mu_q)^2}.$$

This yields 13 means and 13 standard deviations, or 26 scalar features. Other choices include selected percentiles, temporal statistics, or sequence models that consume the full MFCC matrix. The aggregation strategy should be selected before evaluating the test set.

An illustrative implementation is:

```
# mfcc has shape (13, number_of_frames)
mean_mfcc = np.mean(mfcc, axis=1)
std_mfcc = np.std(mfcc, axis=1)
recording_vector = np.concatenate([mean_mfcc, std_mfcc])
```

The code requires `import numpy as np`. If the utterances have different durations, fixed-dimensional statistics can be convenient, although they discard temporal ordering.

9 Why MFCCs may be useful in PD speech research

PD-related speech differences can involve articulation, phonation, and changes in spectral characteristics. MFCCs may capture some of these differences because they summarize the spectral envelope. However, MFCCs are also strongly influenced by phonetic content, speaking task, language, microphone, room acoustics, and recording gain. They should therefore be interpreted as general acoustic descriptors rather than PD-specific biomarkers.

10 Practical considerations and limitations

1. Keep the sampling rate and MFCC parameters consistent across recordings.
2. Consider whether 128 ms windows are appropriate. A shorter window may provide better temporal localization for speech.
3. Decide whether to use magnitude- or power-based Mel energies according to the library defaults and experimental protocol; document the configuration.
4. MFCC values are affected by recording conditions and phonetic content. Use comparable recording tasks where possible.
5. Do not treat the 13 coefficients as independent physiological measurements.
6. If using mean/std aggregation, calculate any scaling or normalization parameters using training data only.
7. If multiple recordings come from the same speaker, use speaker-independent data splits where feasible.

11 Conclusion

MFCCs transform short-time spectral information into a compact cepstral representation. The supplied librosa command returns 13 coefficients for every frame, not a single 13-dimensional vector for the whole recording. For a recording-level classifier, the frame sequence must be aggregated or processed by a temporal model. Parameter selection, recording consistency, and leakage-free evaluation are essential when assessing MFCCs for PD speech classification.

References

[1] S. B. Davis and P. Mermelstein, ‘Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," \emphIEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357-366, 1980. \bibitemlibrosa B. McFee et al., ‘librosa: Audio and music signal analysis in Python,” documentation and software, https://librosa.org/.